Calculated Columns and Math Operators

Lecture Notes · Game DB Junior

In this lesson

- SQL can **process data** before we pull it out of the database
- How to make a **calculated column** (population density)
- The common **math operators**
- How to **rename** a calculated column with **AS**

1. Not only raw data

So far it looks like SQL only pulls out the **raw** information we stored, column by column. But with SQL we can also **transform** or **process** data before we pull it out of the database.

2. Example: population density

Our cities table has the columns **population** and **area**. A useful number to calculate is the **population density** of each city:

population density = population / area
= the number of people per square kilometer in the city

We want to do this calculation for **every row**: Tokyo's 38505000 divided by 8223, Delhi's 28125000 divided by 2240, and so on.

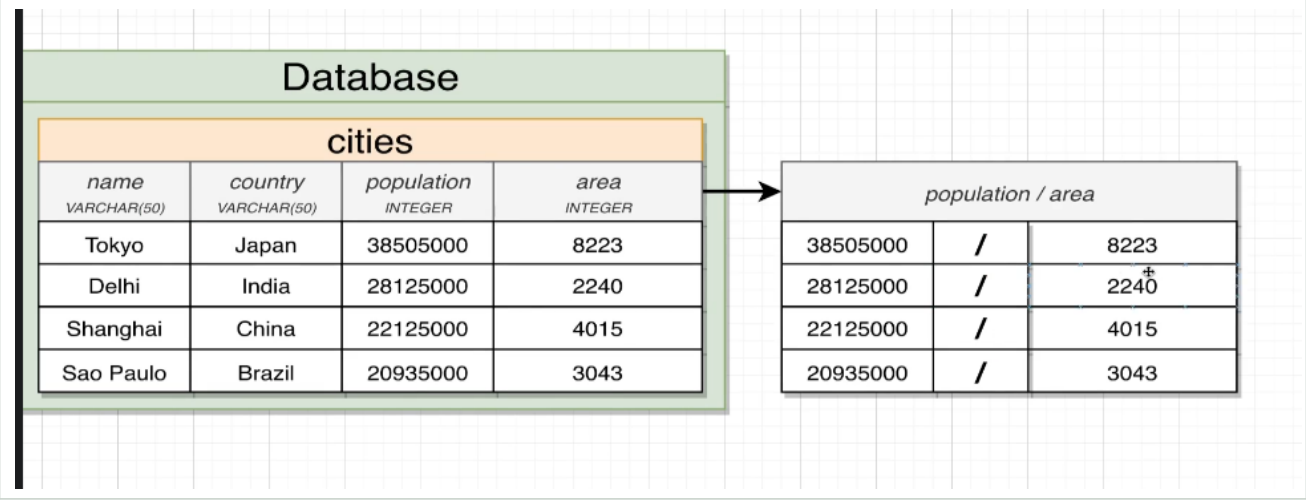


Figure 1 — population / area is calculated for each row

The SQL

```
SELECT name, population / area FROM cities;
```

- First we print each city's **name**.

- Then we calculate each city's **population divided by its area**.
- Think of the result as an **imaginary column**: a temporary calculated column. It is not stored in the table.
- When we run it, we get the name of each city and its population density next to it.

3. Math operators

We are not limited to division. We can use many other math operators:

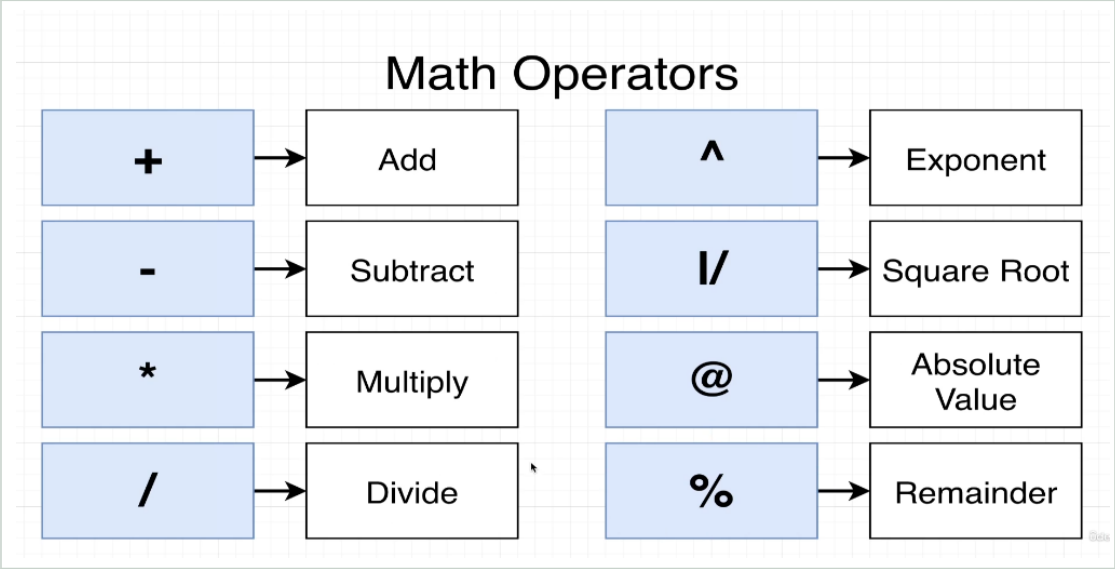


Figure 2 — Math operators

Operator	Meaning	Operator	Meaning
+	Add	^	Exponent
-	Subtract	|/	Square root
*	Multiply	@	Absolute value
/	Divide	%	Remainder

- There are even more math operators, but these are the ones we use most often.
- Out of these, we use the ones on the **left side** (+, -, *, /) the most.

Example: population + area

```
SELECT name, population + area FROM cities;
```

This does not really make sense as a number, but it works. We get a large number for each city.

Example: population * area

```
SELECT name, population * area FROM cities;
```

This gives an **error: integer out of range**. Remember: an INTEGER in Postgres can only go up to about 2.14 billion. Population times area is a number that is too large. Still,

the idea is fine: we **can** multiply two columns.

4. Renaming a calculated column with AS

The result column gets the name **?column?**. Postgres knows we are doing a calculation, but it does not know what to call the result. So it gives this temporary name.

?column? is not a helpful name. It does not tell us what the values mean. To make it clear, we **rename** the calculated column. Right after the calculation, we write **AS** and a new name:

```
SELECT
    name,
    population / area AS population_density
FROM cities;
```

- Now the column name makes much more sense.
- We can use **any name** we want. For example: **AS density** gives a column named density.

5. Running the code (my practice)

Query 1: population density. The result column is named **?column?**.

```
-- Show each city name and its population density
SELECT name, population / area FROM cities;
```

Figure 3 — population / area with a comment

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Data Output | Messages | Notifications

SQL

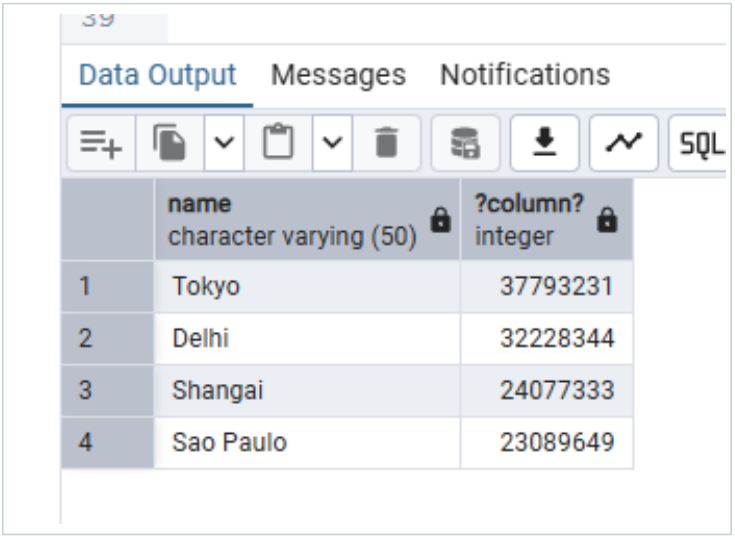
	name character varying (50)	?column? integer
1	Tokyo	4590
2	Delhi	13748
3	Shangai	5555
4	Sao Paulo	6326

Figure 4 — Result: the column name is ?column?

Query 2: population + area.

```sql
-- Show each city name and the sum of population and area
SELECT name, population + area FROM cities;
```

Figure 5 — population + area with a comment



	name character varying (50)	?column? integer
1	Tokyo	37793231
2	Delhi	32228344
3	Shangai	24077333
4	Sao Paulo	23089649

Figure 6 — Result: a large number for each city

Query 3: population * area. The number is too large for INTEGER.

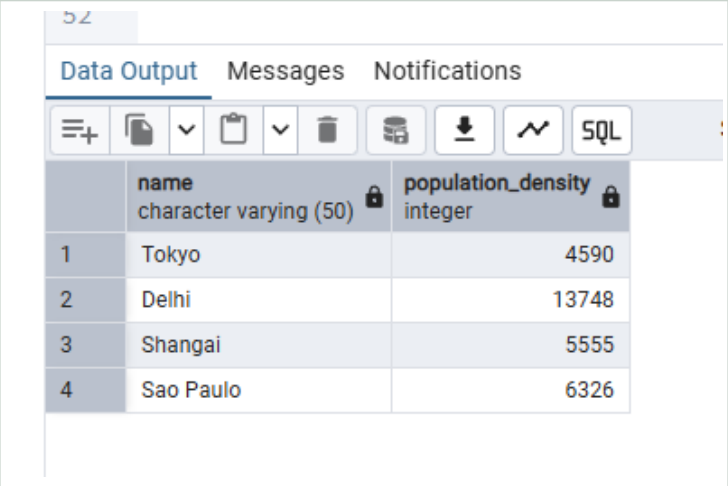
```sql
-- Show each city name and the result of population multiplied by area
SELECT name, population * area FROM cities;
```

*Figure 7 — population * area with a comment*



Figure 8 — ERROR: integer out of range

Query 4: population / area renamed with AS. Now the column is named **population_density**.



	name character varying (50)	population_density integer
1	Tokyo	4590
2	Delhi	13748
3	Shangai	5555
4	Sao Paulo	6326

Figure 9 — Result: the column name is population_density

6. Summary

Idea	Remember
Calculated column	A temporary column made by a calculation. Example: population / area
Math operators	+ - * / are used most. Also ^ |/ @ %
integer out of range	The result is bigger than an INTEGER can hold (about 2.14 billion)
?column?	The temporary name Postgres gives a calculated column
AS new_name	Renames a calculated column. Any name is OK

Sources
Lecture transcript and lecture images (Figures 1–2)
My own practice screenshots (Figures 3–9)